TEAM T081 · SECTION E · CSAI PROJECT 7

Remaining Useful Life Prediction for Turbofan Engines

How many more flights does an engine have left? Engineered degradation features and three classical regressors, evaluated engine by engine on NASA C-MAPSS FD001.



15.36
test RMSE in cycles (selected XGBoost)

7.03
test RMSE within 50 cycles of failure

142
engineered features from 14 sensors

5th · 6th
of 15 FD001 results when scored on labels clipped at 125

Aryan Kumar · Husain Khorakiwala · Navodit Sharma · Sarvesh Srinath · Trishit Swarnakar

github.com/sarveshcore/t081-rul-prediction

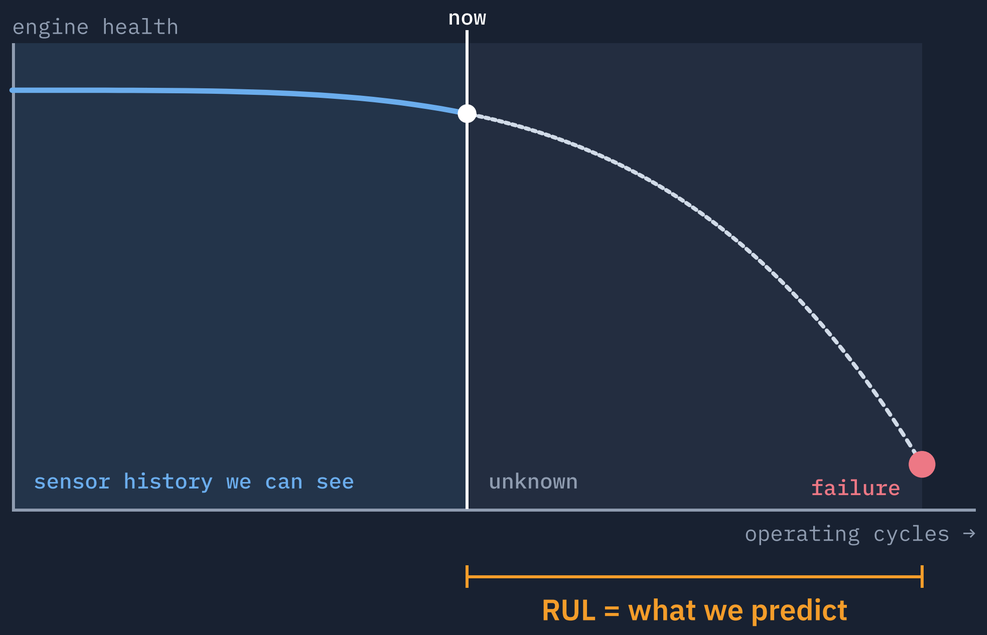
THE PROBLEM

A late prediction is more dangerous than an early one

Remaining useful life (RUL) is the number of operating cycles an engine will still run before it fails. Maintenance today is either on a fixed schedule, which replaces parts that still have life left, or after a fault shows, which risks failure in service. A good RUL estimate lets maintenance happen close to when it is needed.

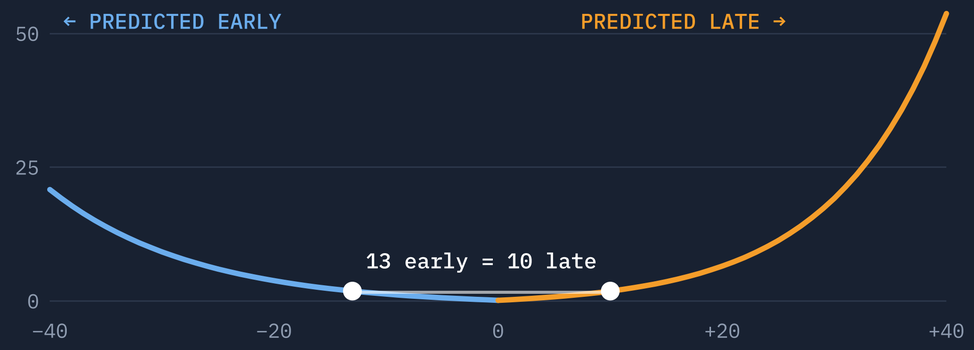
One engine, from new to failure

schematic, not to scale



NASA score penalty for one engine

d = predicted − true RUL; the penalty grows exponentially



PREDICTION OFF BY	TOO EARLY	TOO LATE
10 cycles	1.2	1.7
20 cycles	3.7	6.4
40 cycles	20.7	53.6

Being 10 cycles late costs as much as being 13 cycles early.

We therefore report the NASA score and the error within 50 cycles of failure next to RMSE.

THE DATA · NASA C-MAPSS FD001

The data: 100 engines run to failure, 100 stop early

20,631
training rows, one per engine cycle

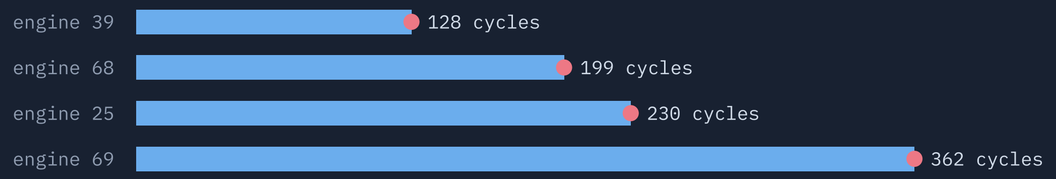
100 + 100
training and test engines

128–362
cycles to failure (median 199)

0
missing values or duplicate rows

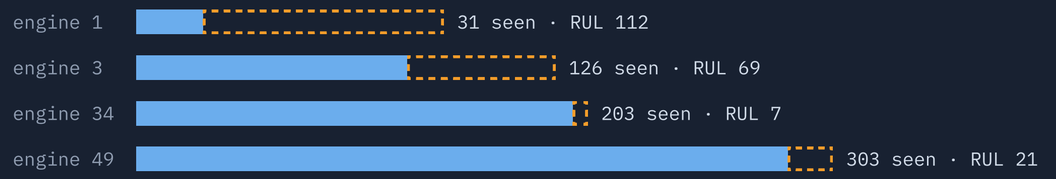
Training: 100 engines, run to failure.

Every cycle gets a label: cycles left until that engine's last cycle. Lifetimes range 128 to 362 cycles.



Test: 100 engines, cut off before failure.

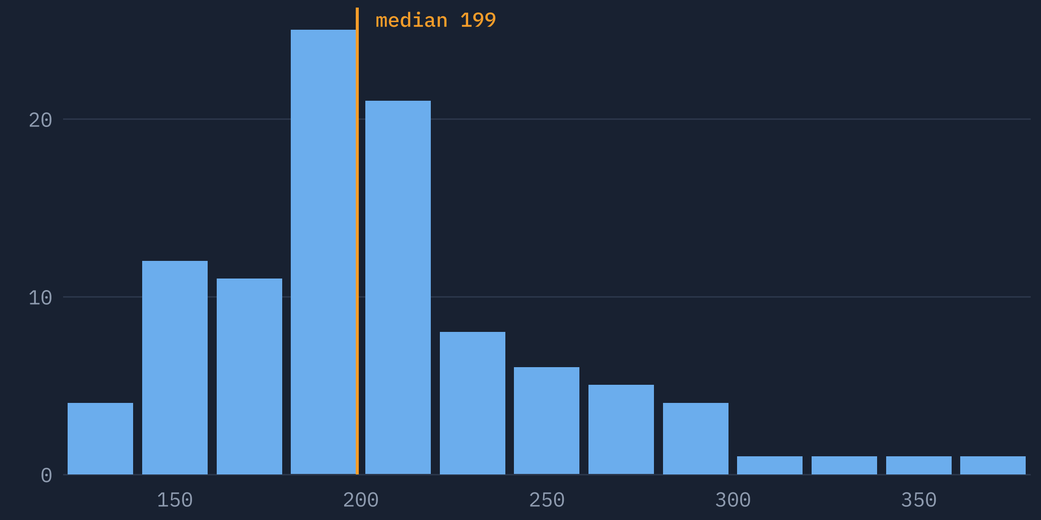
We see 31 to 303 cycles of each; the true RUL at the cut-off (7 to 145 cycles) is given separately and used only for scoring.



cycles observed · failure · true RUL, hidden until scoring

Engine lifetimes

cycles to failure, 100 training engines



Each cycle has 21 sensor readings and 3 operating settings. FD001 has one operating condition and one fault mode.

One RUL prediction per test engine, made at its last observed cycle and scored against the true RUL.

PREPARING THE DATA

Preparing the data: 14 useful sensors and a capped label

Sensor s4 climbs as failure nears

11-cycle trailing mean, five training engines



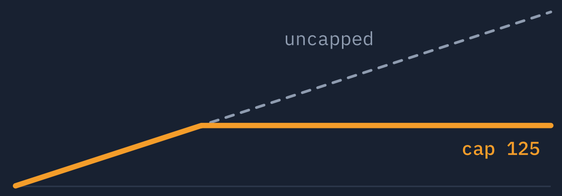
14 of 21 sensors kept

kept in amber

s1	s2	s3	s4	s5	s6	s7
s8	s9	s10	s11	s12	s13	s14
s15	s16	s17	s18	s19	s20	s21

Dropped: s1, s5, s10, s16, s18 and s19 record one value; s6 only 21.60 and 21.61.

Training RUL capped at 125



Changes 38.9% of training labels. Test labels stay uncapped.

What the signal looks like

Sensor s4 (smoothed over 11 cycles) wanders with no clear trend while an engine has more than 125 cycles left, then climbs steadily towards failure. Each engine starts at its own level.

Label cap

Training labels are clipped at 125 cycles, which changes 8,031 of 20,631 labels (38.9%). Test labels stay uncapped.

The cap removes a task the sensors cannot support, telling a healthy engine with 300 cycles left from one with 150, and it roughly halves the error where the wear signal exists.

CV RMSE, SAME MODEL, SAME FOLDS	UNCAPPED TRAINING	CAPPED AT 125
Cycles with true RUL ≤ 125	25.99	**15.09**
All cycles (including those above 125)	**38.26**	47.47

WHAT MAKES IT HARD

Five obstacles, five design decisions

	OBSTACLE	WHAT WE DID
1	**Sensors are noisy and every engine has its own baseline**	Normalise each sensor, then describe each engine by its own recent average, spread and slope
2	**Early in life there is almost no wear signal**	Cap the training label at 125 cycles
3	**Rows of one engine are near copies, so a random row split leaks**	Split by engine with five-fold GroupKFold; refit scalers and the health index inside every fold
4	**Late errors cost more than early ones**	Report the NASA score and error within 50 cycles of failure next to RMSE
5	**One test set of 100 engines**	Freeze the model and every setting by cross-validation, then score the test set once

PIPELINE AND EVALUATION

Pipeline and evaluation: split by engine, never by row

01
Select sensors
14 of 21

→

02
Label
Cycles to failure, capped at 125

→

03
Engineer features
142

→

04
Train learners
Linear · Elastic Net · Random Forest · XGBoost

→

05
Tune and select
5-fold CV by engine

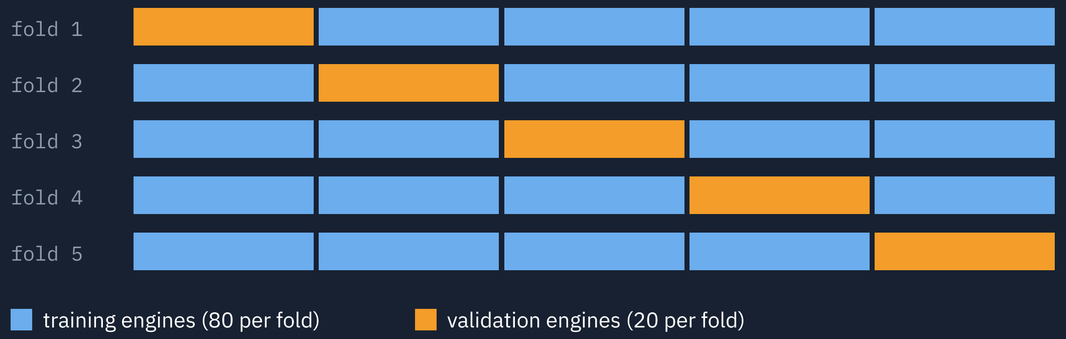
→

06
Test once
100 engines

steps 3 to 5 are refitted inside every fold, on training engines only

Five folds of 20 engines

schematic: GroupKFold assigns whole engines to folds



Every engine is validated exactly once; scalers and the health index are refitted on each fold's 80 training engines.

METRICS

RMSE	Typical miss in cycles; large misses count more
MAE	Average miss in cycles
NASA score	PHM08 penalty summed over engines; lower is better, late costs more
Late-stage RMSE	RMSE on engines with true RUL ≤ 50 (33 of the 100 test engines)

FEATURE ENGINEERING

From 14 noisy sensors to 142 features

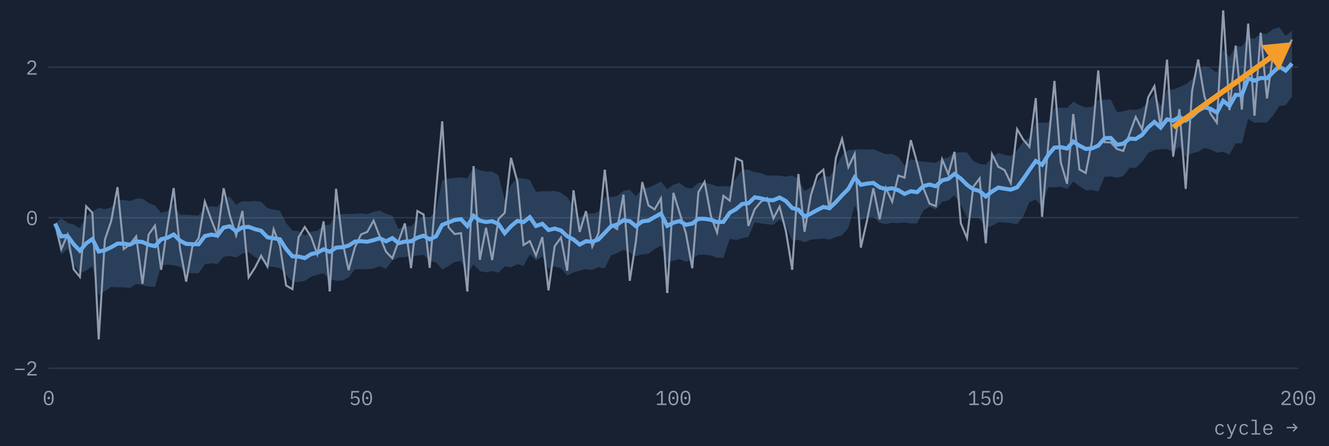
STEP	WHAT IT COMPUTES	FEATURES
Cycle number and raw sensors	The reading itself	1 + 14 = 15
Rolling mean and standard deviation	Level and wobble over the last 5, 10 and 20 cycles, within one engine	14 × 3 × 2 = 84
Rolling slope	How fast each sensor is changing over the same windows	14 × 3 = 42
Health index	One 0-to-1 wear score, regressed from the 14 normalised sensors	1
Total		142

Each sensor is first normalised with training-engine statistics (FD001 has a single operating regime, so this is a plain z-score). Every window stays inside one engine, so no feature mixes two engines.

The level says how worn an engine is; the slope says how fast it is wearing.

One sensor, three views

engine 68, s4 as z-score · line: 10-cycle rolling mean · band: ± 1 rolling std · arrow: 20-cycle slope



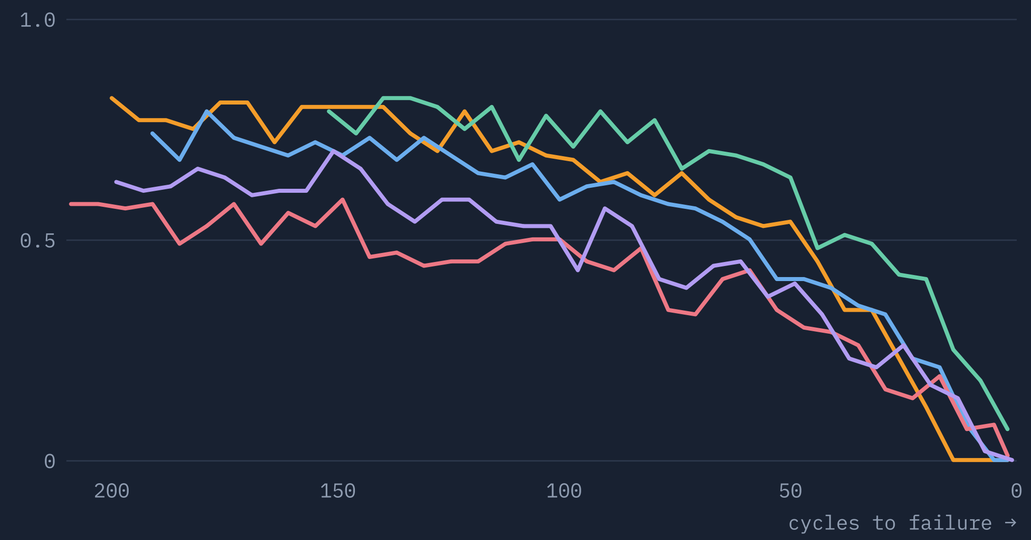
HEALTH INDEX AND ABLATION

The health index and the ablation: slopes help most

Health index. One learned number tracks wear: out-of-fold correlation with capped RUL is **0.87**, against 0.82 for the best single feature (5-cycle rolling mean of s4).

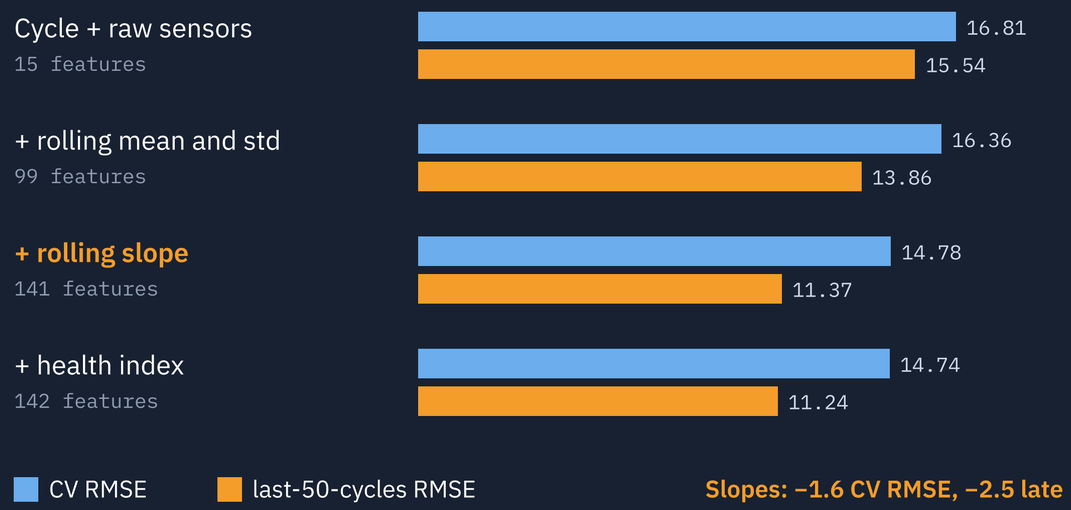
Health index of five training engines

out-of-fold correlation with capped RUL: 0.87 · best single feature: 0.82



Ablation: add one feature family at a time

Random Forest, default settings, 5-fold engine-grouped CV



Slope features cut CV RMSE by 1.6 and last-50-cycles RMSE by 2.5, the largest single step.

MODELS AND TUNING

Four learners, tuned by cross-validation

SIMPLEST → MOST FLEXIBLE

Linear regression

HOW IT WORKS

One weight per feature

CHOSEN SETTING

none

Elastic Net

HOW IT WORKS

Linear regression with L1 + L2 penalties

CHOSEN SETTING

α = 0.001, L1 ratio 0.8

Random Forest

HOW IT WORKS

Many deep trees on bootstrap samples, averaged

CHOSEN SETTING

100 trees, minimum leaf 5, half the features per split

XGBoost selected

HOW IT WORKS

Shallow trees added one at a time, each fixing the remaining errors

CHOSEN SETTING

400 trees, depth 4, learning rate 0.05

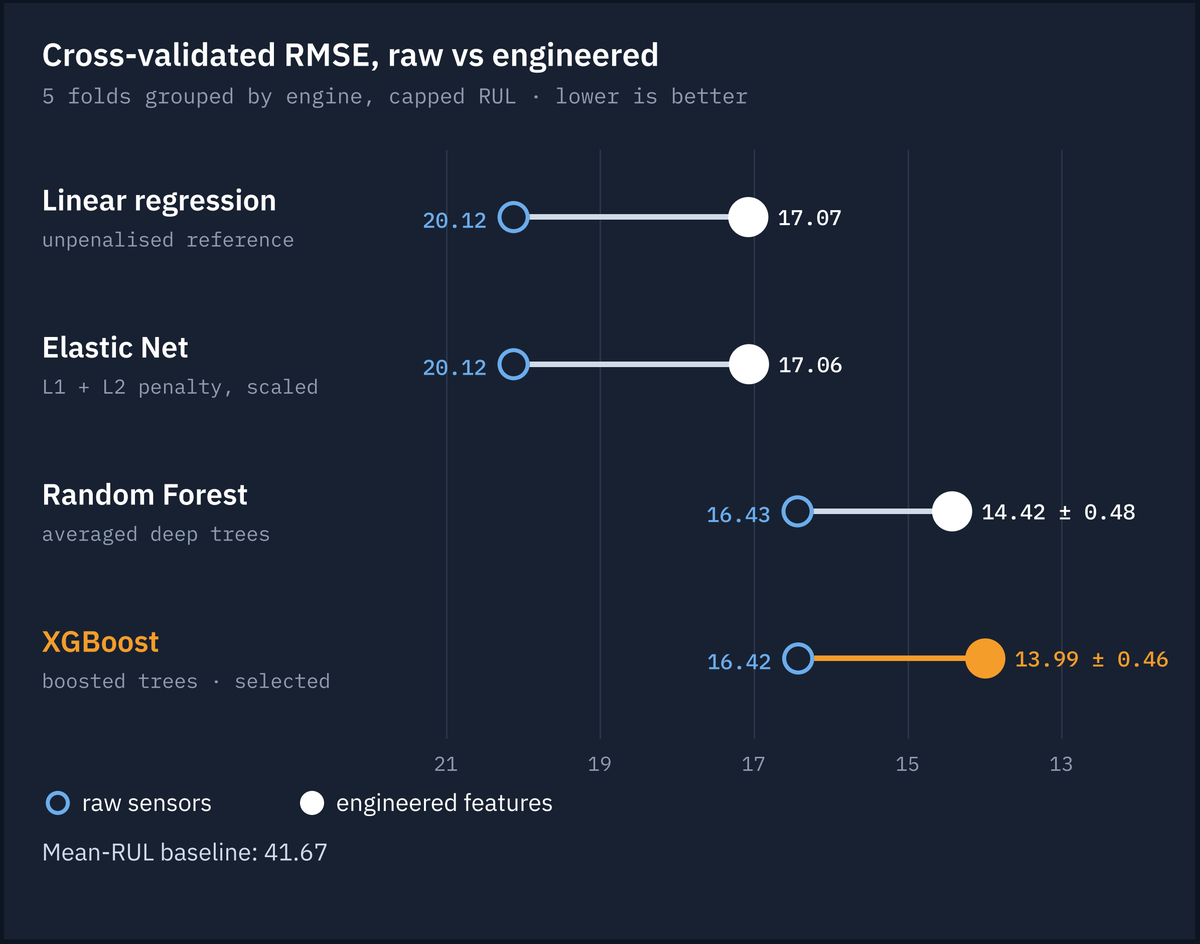
Each model was tuned separately on raw and on engineered features: Elastic Net over a 20-point grid, Random Forest over 6 sampled configurations and XGBoost over 8. The lowest mean CV RMSE wins.

41.67

CV RMSE of a model that always predicts the average training label

RESULTS IN CROSS-VALIDATION

Features help every model, and tree models beat linear ones



Engineered features lower CV RMSE for every model, by **2.0 to 3.1** cycles.

On engineered features, tree models beat linear ones by a further **2.6 to 3.1** cycles.

Elastic Net ties linear regression (17.06 vs 17.07): at the chosen penalty no coefficient is zero. A pure L1 penalty zeroes 38 of 142 coefficients at $\alpha = 0.1$ and 116 at $\alpha = 1$, and the error rises.

XGBoost is selected at **13.99 ± 0.46**, against 14.42 ± 0.48 for Random Forest.

WHAT THE MODELS RELY ON

What the models rely on: the health index

Shuffle one feature group on held-out engines and measure how much the RMSE rises (extra cycles of error).

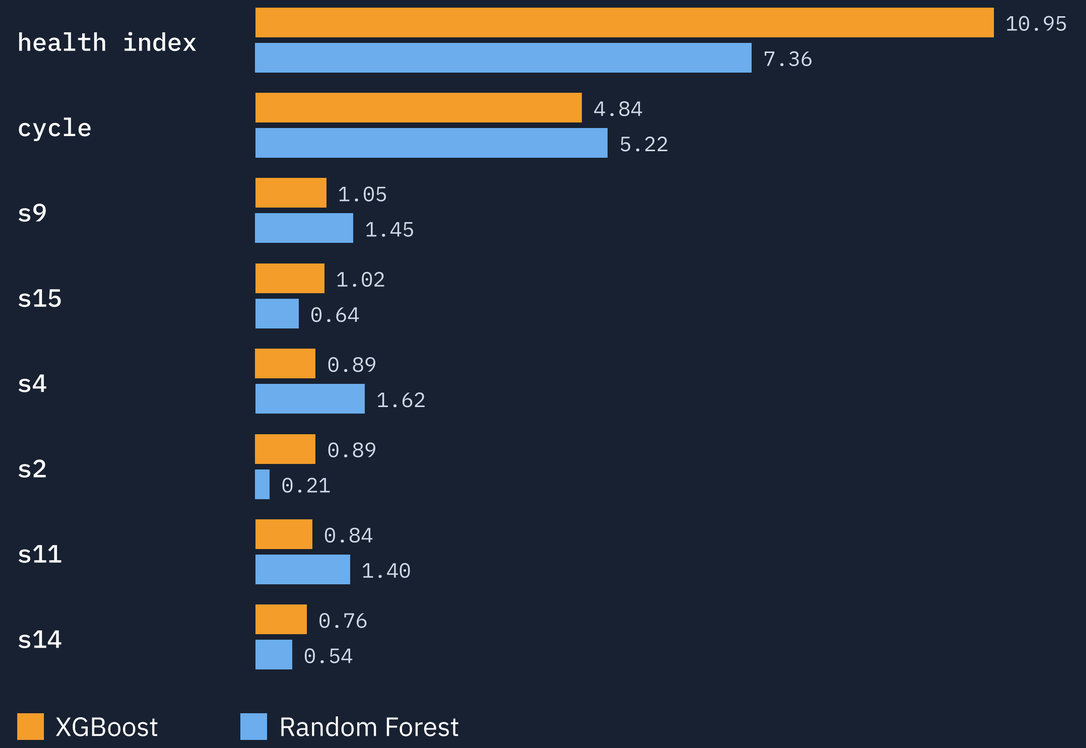
10.95

extra cycles of RMSE when the health index is shuffled for XGBoost (7.36 for Random Forest); nothing else comes close

SHAP on XGBoost agrees: health index first, cycle number second.

Held-out permutation importance

extra RMSE in cycles when a feature group is shuffled · rows ordered by XGBoost



s4 takes 26% of Random Forest's impurity importance but adds only 1.62 cycles when shuffled: impurity rankings overstate correlated inputs.

Each sensor group contains its raw reading, rolling features and slopes.

TEST SET · 100 ENGINES · SCORED ONCE

Test results: 100 engines, scored once

15.36

cycles RMSE, selected XGBoost, all 100 test engines

7.03

cycles RMSE on the 33 engines within 50 cycles of failure

MODEL	RMSE	LAST-50-CYCLES RMSE	NASA SCORE
Mean baseline	43.07	64.09	33,629
Elastic Net	18.27	14.83	522
Random Forest	15.23	7.34	**344**
XGBoost (selected)	15.36	**7.03**	389

Random Forest and XGBoost are statistically tied: a 10,000-sample paired bootstrap puts the 95% interval for the RMSE gap (Random Forest minus XGBoost) at −1.3 to +1.1 cycles. We kept the model that cross-validation chose before the test set was opened.

Engineered features help on test too: XGBoost engineered minus raw is −4.4 to −0.2 cycles, and XGBoost minus Elastic Net is −4.8 to −1.0 (95% intervals).

TEST ERROR BY REMAINING LIFE

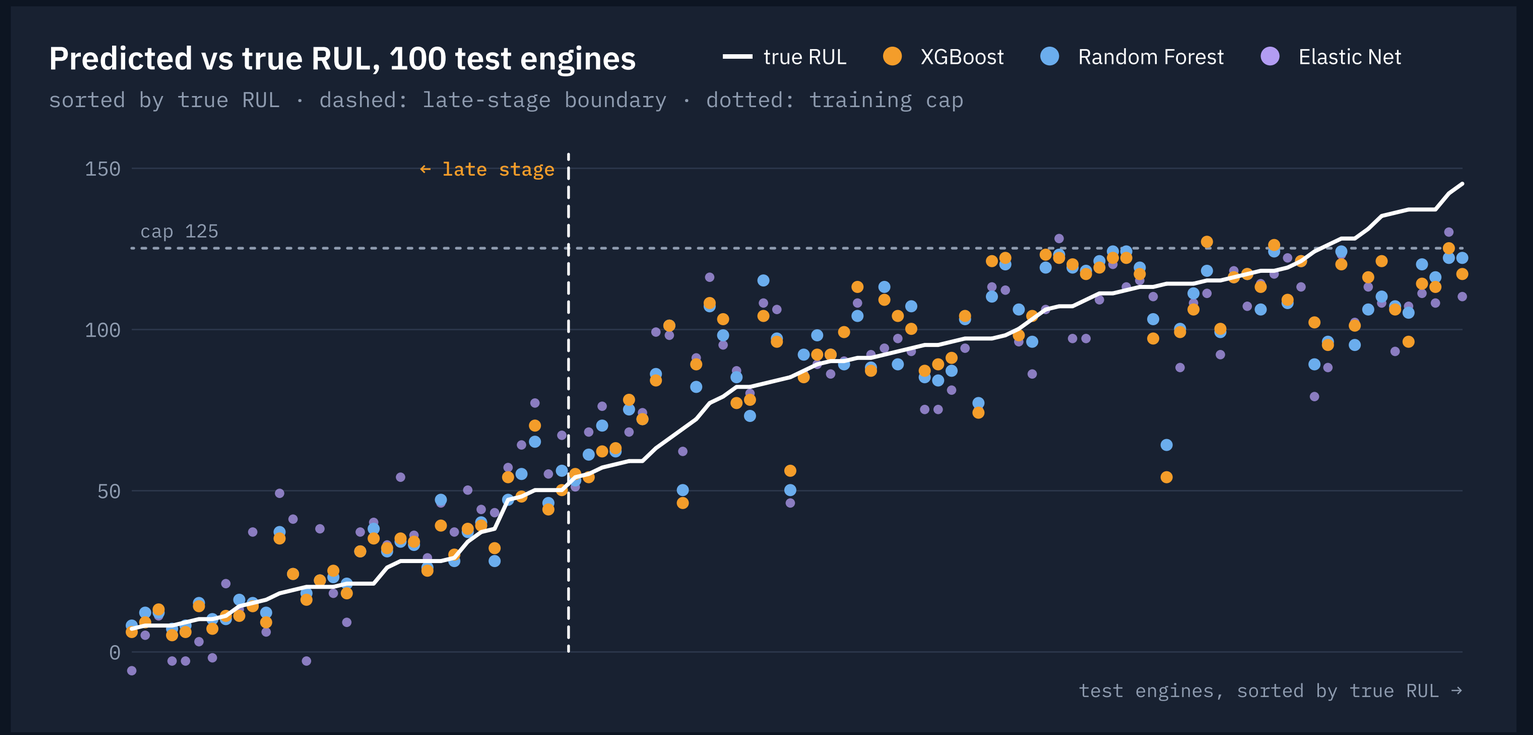
Accurate near failure, limited far from it

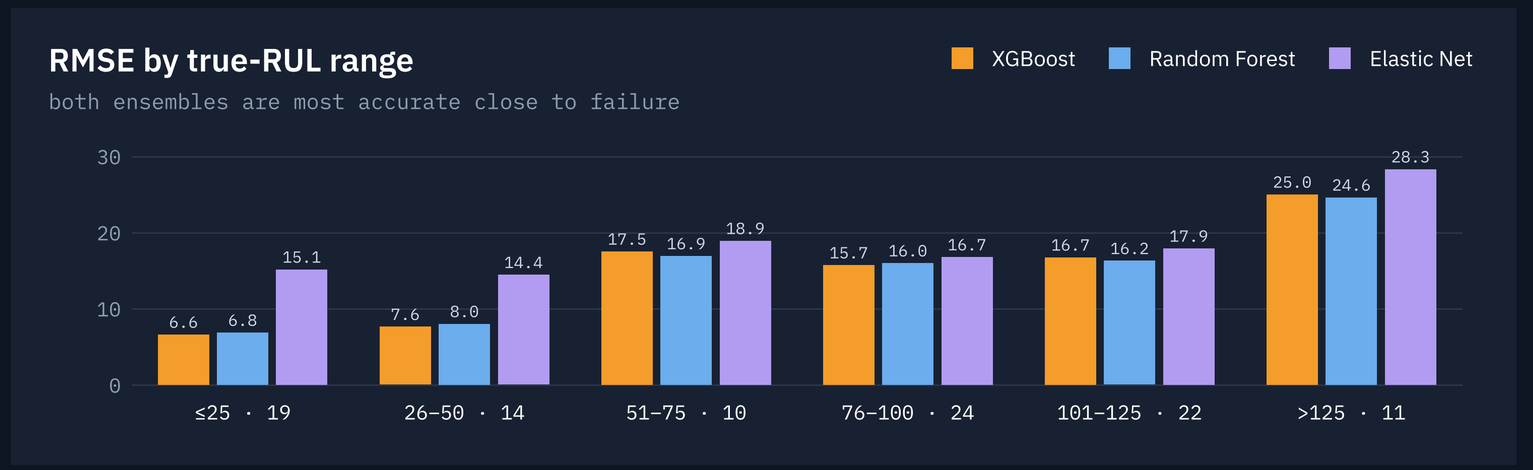
RMSE on the test set by true remaining life; the number after each range is its engine count.

Both ensembles are about 7 cycles off within 50 cycles of failure, where maintenance is decided; Elastic Net is about twice as far off.

In the 51–75 range every model predicts too much life left (median error +9 for XGBoost, +11 for Random Forest, +14 for Elastic Net).

The 11 engines with more than 125 cycles left are always underestimated, because the training label is capped at 125.





PUBLISHED COMPARISON

Close to deep models with hand-built features

Test RMSE against labels clipped at 125, the convention in published papers (published values from Ragab et al. 2020, Table III; lower is better).

RANK	METHOD	RMSE	NASA SCORE
1	1D CNN (Li et al. 2018)	12.61	274
2	ATS2S attention (Ragab et al. 2020)	12.63	243
3	HDNN (Al-Dulaimi et al. 2019)	13.02	245
4	BLCNN (Liu et al. 2019)	13.18	302
5	**Random Forest (ours)**	**14.02**	**310**
6	**XGBoost (ours)**	**14.08**	**347**
7	CNN-LSTM (Wu et al. 2019)	14.40	290
8	BiLSTM-ED (Yu et al. 2019)	14.74	273
9	LSTMBS (Liao et al. 2018)	14.89	481
10	MODBNE (Zhang et al. 2017)	15.04	334
11	Gradient boosting (Zhang et al. 2017)	15.67	474
12	Deep LSTM (Zheng et al. 2017)	16.14	338
13	**Elastic Net (ours)**	**17.01**	**469**
14	Random Forest (Zhang et al. 2017)	17.91	480
15	2D CNN (Babu et al. 2016)	18.45	1,287

published deep · published trees · ours

These are the same predictions as slide 12, scored against clipped test labels: XGBoost 15.36 becomes 14.08.

Our Random Forest trails the four leading deep models by 0.8 to 1.4 cycles (XGBoost by 0.9 to 1.5) and is ahead of six deep models and both published tree models.

On the NASA score, six published models beat our Random Forest (310) and eight beat our XGBoost (347).

FINDINGS AND LIMITS

What this evaluation shows, and its limits

2.0–3.1

Engineered features help every model, lowering CV RMSE by 2.0 to 3.1 cycles; rolling slopes were the largest single step.

2.6–3.1

Tree models beat linear ones by a further 2.6 to 3.1 cycles; Random Forest and XGBoost are tied on the test set.

10.95

A single learned wear score carries most of the signal: shuffling the health index adds about 11 cycles of error.

7.03

Error is smallest where it matters: 7.03 cycles RMSE within 50 cycles of failure, against 15.36 overall.

LIMITS OF THIS EVALUATION

- One operating condition and one fault mode (FD001).
- 100 test engines with one prediction each, so intervals are wide; 33 are in the late stage.
- Engines with more than 125 cycles left are always underestimated because of the cap.
- Features look back at most 20 cycles, with windows chosen by hand.
- C-MAPSS is simulated and published protocols differ, so the ranking is approximate.

PHASE 2 · NEXT

Phase 2: let the model read the whole sensor history

2A · Sequence models

LSTM, temporal CNN and Transformer encoder on fixed-length windows of the raw sensors, replacing our hand-built 5-, 10- and 20-cycle summaries.

2B · Hybrid and ablation

Feed our engineered features into the sequence model, and ablate each part, to test whether hand-built features still help once the history is learned.

ALREADY FOUND · TREND MATTERS

16.36 → 14.78

Adding rolling slopes was our largest gain (CV RMSE, last-50-cycles RMSE 13.86 → 11.37). Our features look back at most 20 cycles; a sequence model can read all of them.

BAR TO BEAT

15.36 RMSE

Selected XGBoost on the 100 test engines; 7.03 within 50 cycles of failure, 14.08 on labels clipped at 125. Published deep models reach 12.61 to 13.18.

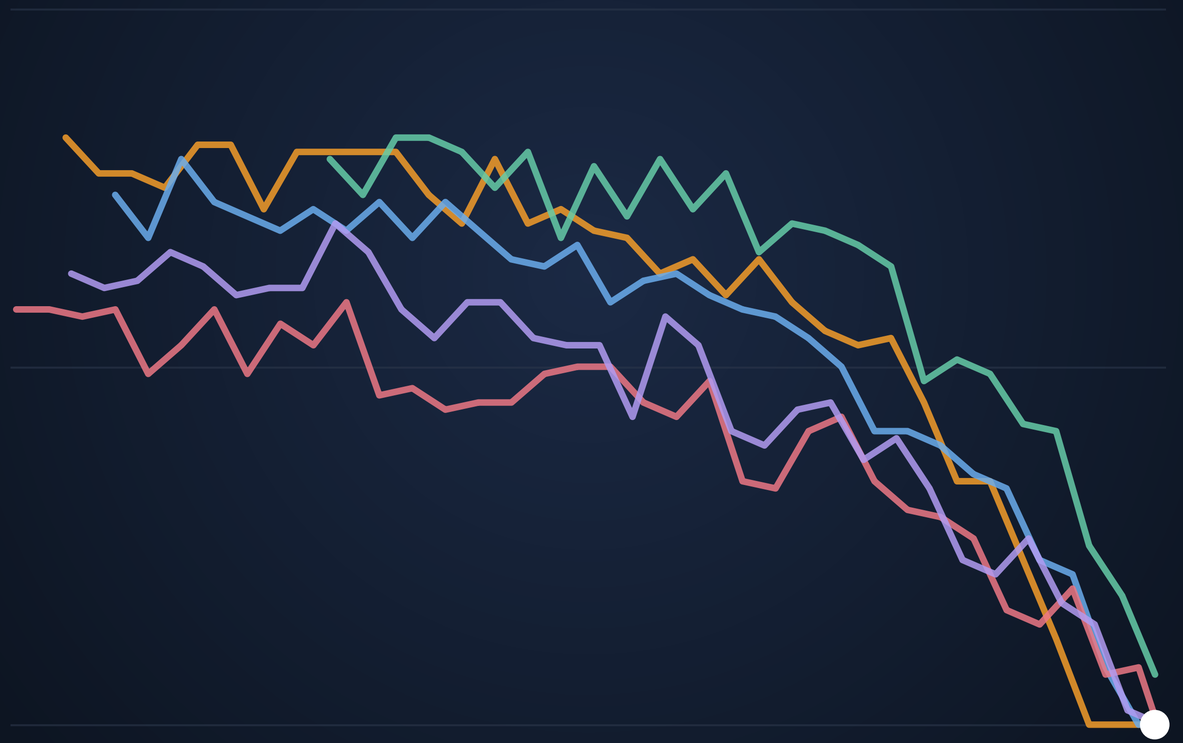
Same protocol: FD001 · five folds grouped by engine · training RUL capped at 125 · test set scored once · RMSE, NASA score, late-stage RMSE

THANK YOU

Thank you

Questions?

Same engines, same folds, same metrics: every number in this talk can be rerun from the repository.



health index of five training engines · cycles to failure →

Aryan Kumar · Husain Khorakiwala · Navodit Sharma · Sarvesh Srinath · Trishit Swarnakar

Team T081 · Section E · CSAI Project 7

github.com/sarveshcore/t081-rul-prediction